Tessera Integrated HSM ASIC · Chip Design Specification

Source: Tessera Chip Requirements Specification (CRD) v0.5.0 · 315 requirements · all descriptions are verbatim CRD text · ~~strikethrough~~ as in the CRD

github.com/TesseraHSM/tessera-specs @ bcdb753 · 2026-10-29

ASIC at a Glance

block view (not a floorplan) · values from the CRD with requirement IDs · → section with full text

HOST · VMs / TEEs — ↑↓ PCIe Gen6 · IDE / TDISP

Tessera die · Physical Security → §6

Certification FIPS 140-3 level 3 · full chip boundary `Physical Security`
Sensors Tamper Shield · Temperature · Frequency and Voltage monitors `Physical Security`
Intrusion 2x tamper pins · battery backed RTC domain `CINTR-1 / HW-RTC-1` **Debug lock** fuse-based `HW-DBG-3`
Separation IDE · SP · key loader physically separate `Physical Security` → §6

Host Interface → §1–2

PCIe	**Gen6**	BASE-1
Lanes	**4**	BASE-2
Bifurcation	**2×2 / 1×4**	BASE-3
VFs	**512 VF per PF**	BASE-15
IDE	**AES 256 GCM**	SEC-PCIE-1
IDE streams	**≥2 Selective + Link**	SEC-PCIE-7
TEE I/O	**TDISP**	SEC-PCIE-2
Attestation	**SPDM 1.5**	ROT-8
DMA	**PRP and SGL in hardware**	GDMA-1

Control Core Cluster → §3

Cores	**At least 4**	CPU-1
Class	**ARM M7/RISC-V E6**	CPU-2
Clock	**≥ 1 GHz / ≥ 900MHz**	CPU-3 / BASE-10

Security Processor · RoT → §6

RoT	**Caliptra 2.2**	ROT-1
Identity	**PUF**	ROT-14
Boot	**OCP streaming boot**	ROT-3
Anti-rollback	**≥ 32 counts**	ROT-11

GSRAM · Key Vault → §3–4

GSRAM	**≥ 8MB**	MEM-2
Key Vault	**4MB**	MEM-2
ECC	**SECDED**	MEM-3
Apertures	**≥ 16**	MEM-4

Key Mover → §4

Rate	**400k keys / s**	CRYPT-KM-1
Cipher	**AES-GCM · 256 bit key**	CRYPT-KM-2
Throughput	**3.2 GB/sec**	CRYPT-AES-6

Cryptographic Accelerator Pools → §5, §13–14

PQC	**ML-DSA 44 / 65 / 87 · ML-KEM 512 / 768 / 1024**	CRYPT-PQC-6/7
RSA	**2048 / 3072 / 4096**	CRYPT-RSA-3
ECC	**P-256 · P-256K · P-384 · P-521**	CRYPT-ECC-1..4
AES	**128 / 192 / 256 · ECB CBC CTR CFB OFB GCM KW**	CRYPT-AES-1..9
Hash	**SHA1 · SHA2 · SHA3 · SHAKE**	CRYPT-HASH-1..4
RNG	**NIST 800-90C**	CRYPT-RNG-1

Host Memory Hasher → §5

Rate	**≥ 3.2GB/sec**	CRYPT-HMH-2
Directives	**≥ 32 pending**	CRYPT-HMH-5

Peripheral IO · Debug → §7–8

Flash	**2x QSPI**	HW-QSPI-1
Mgmt	**3x I2C/I3C**	HW-I3C-1
USB	**USB2 device**	HW-USB2-10
GPIO	**12**	HW-GPIO-1
Debug	**JTAG / SWD · UART**	HW-DBG-1/2

Performance targets → §13 (all 33)

ECC Sign p-256	ML-DSA-44	ML-KEM-512	RSA Sign 2048 CRT	AES 256 CBC/ECB/CTR/OFB/GCM	SHA 2-256
200,000 OPS	**200,000** OPS	**200,000** OPS	**100,000** OPS	**1,600** MB/s	**3,100** MB/s
PERF-7	PERF-32	PERF-35	PERF-1	PERF-17	PERF-25

Package · Power · Test · Reliability → §10–12

TDP	Package	Type	Tj	Scan	MBIST	SER
≤ 12 W	**14 mm x 14 mm**	**Flip-Chip BGA (SHOULD)**	**0°C to 105°C**	**> 99% stuck-at**	**100% coverage**	**SDC 50 FIT · DUE 250 FIT**
PWR-2	PKG-1	PKG-2	THERM-1	MFG-ATE-1	MFG-ATE-2	REL-SER-1/2

Labels and block grouping by the editor; every value is CRD text (shortened where marked with "/" or "·") with its requirement ID. CPU-3 and BASE-10 state different clock rates; both are shown.

Values are CRD v0.5.0 (draft) requirements, not measured results. This sheet covers 315 of the 375 CRD requirements that bear directly on chip design; the rest (firmware logging, telemetry, live migration, etc.) are in the CRD.

Architecture Principles (CRD text)

Tessera SHALL be certified to FIPS 140-3 level 3 at the full chip boundary. Tessera's cryptographic acceleration blocks, memories, and buffers storing keys in plain text SHALL have the following tamper defense mechanisms: 1. Tamper Shield 2. Temperature Sensor 3. Multi Register and Multi bit Registers with parity protection on memory protection configuration. 4. Security sensitive registers shall be multi-bit. Example: a single bit should not represent debug enabled. 5. Frequency and Voltage monitors

The PCIe IDE, Security processor, and key loader encryption functions are invisible to the tenant and exist for the device's secure functioning, and MUST exist as physically separate entities.

The PQC, RSA/ECC, host memory hasher, and utility pool functions cover operations needed for the user-visible function of the device.

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1 PCIe / Host Interface

31 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
BASE-1	**PCIe Standard:**	Gen6
BASE-2	**Number of Lanes:**	4
BASE-3	**Bifurcation:**	2×2 with multi-host support, 1×4 single host
BASE-4	**NVMe Version**	2.0 as the baseline, on PF0 only. Commands will be vendor specific.
BASE-5	**NVMe CMD descriptor format**	Both PRP and SGL modes supported. HW acceleration of PRP required, SGL required
BASE-6	**NVMe CMD max IO size**	16 MiB
BASE-7	**NVMe SQs**	1536+8 on PF0
BASE-8	**NVMe CQs**	1536+8 on PF0
BASE-9	**Flexible allocation of SQ/CQ pairs to VF**	Required. Shall be able to flexibly allocate # of SQ/CQ pairs per VF from 3 to 1536
BASE-14	**Number of Physical Functions (PF):**	2 per EP. In single port mode, this is a total of 2PFs. For dual port (bifurcated 2×2 mode) 2 PFs per port, total of 4.
BASE-15	**Number of Virtual Functions (VF):**	Single port (1×4): PF0 512 VF, PF1 512 VF per port · Dual port (2×2): PF0 256 VF, PF1 256 VF per port
BASE-16	**Interrupt support**	MSI-X
BASE-17	**MSI-X Interrupt vector**	≥ 1544 (1536+ 8) for PF0 ≥ 2052 (2048+ 4) for PF1 MSI-X Interrupt vectors must be freely assignable to VMs, to accommodate a range of VM sizes.
BASE-20	**Max payload size (MPS) supported**	512, 1024 bytes
BASE-21	**Max Read Request Size support**	4KB
BASE-22	**PCIe Tag Support**	10-bit tag support is required. 14-bit tag is desired
BASE-23	**Multi Host**	Supported (including separate refclk, PERST, Link PM, isolated data path)
BASE-25	**Function Level Reset**	Supported
BASE-26	**Virtual Function level Reset**	Supported
BASE-27	**Advanced Error Reporting**	Supported
BASE-28	**PCIe FLIT logging support**	Required. FBER measurements and lane error status

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
BASE-32	**L0p support**	Required
BASE-33	**Power Management**	D0 and D3 (exit latency ~200ms), Link L0, L1.1 and L1.2 -ASPM (exit latency < 15us) and L2 (or equivalent low power state) Dynamic Link Width Control
BASE-36	**Separate REFCLK (SRIS/SRNS)**	To support HPAR/multi-host.
BASE-37	**Hot Plug/Surprise Removal**	Support hot plug/surprise removal of one host in a Multi Host configuration without impacting functionality/performance of the other connected host. Enhanced Downstream Port Containment support shall be required.
BASE-38	**MCTP Base Specification 1.3.1**	MCTP over PCIe 1.2.0: Support for telemetry & FW deployments MCTP over I3C 1.0.0: Support for telemetry (incl FW logs). Required on all I3C interface.
SCAL-1	**Dual host support**	Tessera shall support dual hosts by bifurcating its PCIe x4 interface into two independent PCIe x2 interfaces.
SCAL-2	**TDISP in dual host mode**	In a dual host mode, Tessera SHALL support TDISP and IDE over each x2 interface.
SCAL-3	**PF/VF in dual host mode.**	In dual host mode, each of 2 host interfaces SHALL support 2 PFs each with 256 VFs.
SCAL-4	**Crypto apportioning**	In dual host mode, each host interface SHALL support each host using half of the crypto performance of the whole part.
SCAL-5	**Asymmetric crypto apportioning**	In dual host mode, each host interface SHALL support a single host using all of the crypto performance of the whole part.

2 PCIe Security (IDE / TDISP / SPDM)

21 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
SEC-PCIE-1	**GCM encryption**	The ASIC must support AES 256 GCM with 96bit TAG as per PCIe Gen6 IDE specification.
SEC-PCIE-2	**TDISP**	The ASIC must support PCIe TEE Device Interface Security Protocol (TDISP)
SEC-PCIE-3	**Key agreement**	AES Key agreement algorithm shall follow NIST Special Publication 800-133 Recommendation for Cryptographic Key Generation, and (pending approvals in standard bodies) for key establishment.
SEC-PCIE-4	**Attestation**	Attestation of the PCIe Endpoint device shall support Identity and device measurement claims according to SPDM 1.5 pending approvals in the standard bodies.
SEC-PCIE-5	**Tessera identity**	Tessera hall support authentication chained to Caliptra's Device Identity Certificate.
SEC-PCIE-6	**PCIe IDE keys private**	Tessera will not expose the PCIe IDE keys to any entity other than the PCIe IP itself and the security processor.
SEC-PCIE-7	**Selective and Link support**	Device shall support at least two Selective IDE streams when TDISP disabled and 1 Selective IDE stream when TDISP enabled. Link IDE shall be supported.
SEC-PCIE-8	**Key sets**	Each IDE stream shall support storage of at least two key sets for encrypted TLPs.
SEC-PCIE-FEATURES-1	**CMA BaseAsymAlgo**	Tessera SHALL have hardware support for the following sign and verify ("BaseAsymAlgo") operations specified within Component measurement and authentication (CMA/SPDM1.4): RSASSA-3072 ECDSA P-256 ECDSA P-384 ECDSA P-521
SEC-PCIE-FEATURES-2	**SPDM PQC support**	Tessera SHALL have hardware support for the following sign and verify ("PqcAsymAlgo") operations specified within Component measurement and authentication (CMA/SPDM1.4): ML-DSA-44, ML-DSA-65, ML-DSA-87
SEC-PCIE-FEATURES-3	**SPDM Hashing support**	Tessera SHALL have hardware support for the following hashing ("BaseHashAlgo") operations specified within CMA: SHA_256, SHA_384,SHA_512.
SEC-PCIE-FEATURES-4	**SPDM RNG support**	Tessera SHALL have access to an on-chip random number generator per NIST guidelines for secured SPDM session establishment.
SEC-PCIE-FEATURES-5	**SPDM over DOE**	SPDM and Secured SPDM messages for CMA SHALL be supported in-band over PCIe via DOE.
SEC-PCIE-FEATURES-6	**Encrypted SPDM**	Secured SPDM messages SHALL be encrypted and authenticated using the AES-256 GCM option.
SEC-PCIE-FEATURES-7	**SPDM key exchange methods**	Secured SPDM session key exchange SHALL support both options: pre-shared secret and SIGMA option (based on ephemeral Diffie-Hellman and digital signatures).

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
SEC-PCIE-FEATURES-8	**Hardware accelerated SPDM**	Tessera SHALL have hardware assisted support for encryption and decryption of secured SPDM messages transmitted via DOE using Caliptra features.
SEC-PCIE-FEATURES-9	**DOE availability**	Each upstream port capable of IDE SHALL support one instance of DOE hardware.
SEC-PCIE-FEATURES-10	**DOE buffering**	Each DOE instance shall allow buffering of messages with at least ~~16 KiB~~ 4kB capacity in each direction.
SEC-PCIE-FEATURES-11	**Doorbell stride**	Doorbells SHALL be configurable at 2kB stride to allow the doorbells for a single queue pair to appear exclusively in a single 4kB page.
CRYPT-IDE-1	**Direct Release**	AES GCM PCI IDE keys MUST be released to the PCIe controller directly from the security processor that terminates SPDM.
CRYPT-IDE-2	**Periodic self-test**	The PCIe IP MUST be capable of a FIPS-140 compliant self-test at startup and periodically during operation without interrupting or disabling any existing sessions.

3 Processors, Memory and Data Movement

19 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
BASE-10	**Processor Complex size**	At least 4x M7 or E6 class cores, at least 900MHz. At least 64kB instruction cache and 64kB data cache per core.
BASE-11	**GSRAM size**	At least 8MB
BASE-12	**Key Vault Size**	Configurable partition of GSRAM, that will contain obfuscated key material; operation with minimum KV size of 4MB MUST be supported.
CPU-1	**Core cluster**	Quantity: At least 4 cores · Tessera SHALL have a general purpose core cluster, sufficient to support administrative and asymmetric throughput for HSM flows
CPU-2	**Core cluster per-core perf**	Quantity: ARM M7/RISC-V E6 · Each core SHALL perform at least that of an ARM M7 or RISC-V E6 class CPU
CPU-3	**Core cluster clock rate**	Quantity: At least 1 GHz · Each core SHALL have at least 1GHz clock rate.
CPU-4	**Core cluster cache size**	Quantity: At least 64kB instruction cache, 64kB data cache · Each core SHALL have at least 64kB instruction cache, 64kB data cache.
CPU-5	**Core cluster working store**	Quantity: GSRAM · The core cluster SHALL have its code and data storage within a configurable aperture in shared GSRAM.
MEM-1	**GSRAM**	Tessera SHALL have a unified GSRAM, containing core processor cluster code and data and key vault data.
MEM-2	**GSRAM size**	Quantity: At least 8MB AND enough to support 4MB KV. · Tessera's GSRAM SHALL be sized adequately for full rate full function operation (see FW-SYS-1) with at least 4MB of key vault space used to hold plaintext keys in pinned mode. Independently of this functional requirement, the GSRAM SHALL be at least 8MB in size. (including provision for FW-SYS-2 and FW-SYS-3)
MEM-3	**GSRAM integrity**	Quantity: At least SECDED · The GSRAM SHALL have at least SECDED ECC.
MEM-4	**GSRAM apertures**	Quantity: At least 16 · The GSRAM SHALL be configurable into at least 16 distinct contiguous regions, each with independent configuration set by the SP on boot indicating address range and which bus initiators can read or write.
MEM-5	**GSRAM range config**	Memory aperture ranges SHALL be configured at boot by the security processor and attestable during runtime.
MEM-6	**GSRAM aperture granularity**	Quantity: At most 64kB · Memory aperture ranges SHALL be configurable in a granularity at most 64kB
GDMA-1	**PRP and SGL support**	PRP and SGL decoding and fetching of pages for bulk transfers SHALL be offloaded to hardware engines and not be performed by CPU cores.
GDMA-2	**Priority levels**	The GDMA engine SHALL support enough priority levels to ensure every user of PCIe bandwidth (core cluster, host memory hasher) can receive a minimum bandwidth.
FW-SYS-1	**Perf measurement boundary**	Tessera MUST meet the performance requirements for throughput, transactions and latency as specified in this CRD with the production firmware load and driver interface, as measured from the host driver, and including all runtime checks needed to operate in FIPS 140-3 compliant mode. The architecture MUST provide enough application processors and memory to meet this requirement.
FW-SYS-2	**Scratch memory**	Scratch memory MUST be available for the purposes of but not limited to logging, collecting telemetry data, and impactless firmware update. The minimum size of scratch memory is *max*(640 KiB, *max*(TCM size of any Tessera processor core))

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
FW-SYS-3	**Headroom**	Tessera's architecture MUST provide a justifiable 20% headroom in core cluster cycles and GSRAM capacity as delivered, as compared to estimated load when processing hybrid ML-DSA-2 and ECC 256p signing at full rate. This is to allow future firmware features to be added during the device's lifetime and is not intended to be a separate processing partition.

4 Key Architecture (Pinned / Offloaded / Key Mover)

27 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
KEY-1	**Restricted plaintext storage**	Keys in plaintext format SHALL be stored only in dedicated key store locations that are configurably protected from general firmware access.
KEY-2	**Write restriction**	Write access to key store locations SHALL be restricted to specified HW components or by FW executing on specified cores.
KEY-3	**Read restriction**	Read access to key store locations SHALL be restricted to specified HW components or by FW executing on specified cores.
KEY-4	**Key storage models**	Tessera SHALL support a mode of operation where the plaintext keys for any individual operation is already resident in the device's key vault called 'pinned mode' and another mode of operation where the keying material used for any operation is provided in encrypted form along with the operation and decrypted into the key vault for that one use, called 'offloaded mode'.
KEY-OFF-1	**Offloaded model**	Tessera SHALL support a mode of operation where the keys for every asymmetric operation are provided in encrypted 'sealed' form along with the operation and its arguments.
KEY-OFF-2	**Offload encryption**	Offloaded keys SHALL be AES-GCM encrypted.
KEY-OFF-3	**Sealing key size**	Quantity: 256 bits · The sealing encryption key size SHALL be 256 bits.
KEY-OFF-4	**Sealing key IV**	Quantity: 96 bit counter deterministic · The AES GCM IV SHALL be 96 bits counter deterministic, 0 at initialization.
KEY-OFF-5	**VM zeroization**	The ephemeral sealing keys associated with any VM SHALL be zeroized on any VM disconnect.
KEY-OFF-6	**Session zeroization**	The ephemeral sealing key associated with any session SHALL be zeroized when that session is closed.
KEY-OFF-7	**Dedicated hardware**	Tessera SHALL incorporate a 'key mover' handling AES-GCM decryption and validation of sealed keys.
KEY-OFF-8	**Sealing key derivation**	The AES-GCM sealing keys SHALL be derived using an APPROVED KDF incorporating at least 256 bits from an APPROVED DRBG, a 256 bit nonce from the tenant VM, and a 256 bit value derived from the individual Tessera's device identity.
KEY-OFF-9	**Sealing key derivation time**	The sealing key for an individual session SHALL NOT be derived before the session has been opened.
KEY-OFF-10	**Sealing key validation**	Source, usage and type of key expected SHALL be validated ~~with the authenticated data in the offloaded key~~ and a distinct error SHALL be indicated on any failure.
KEY-OFF-11	**GCM validation**	If AES-GCM authentication is not successful a distinct error SHALL be indicated.
KEY-PIN-1	**Pinned KV**	Tessera SHALL support configuring at boot a Key Vault region of the GSRAM for storing pinned keys.
KEY-PIN-2	**KV Max size >= 4MB**	Tessera SHALL support full performance while providing 8kB of pinned key vault space for each of 512 tenants.
KEY-MGT-6	**Internal transport of plaintext keys.**	Keys SHOULD be moved from key registers to hardware acceleration blocks without control core firmware having visibility to the plain text key values.
KEY-MGT-7	**Zeroization**	Tessera's architecture MUST explicitly identify any memories, GSRAM or otherwise, inside or outside accelerators that can hold obfuscated plaintext keys and support explicit mechanisms for firmware to zeroize individual keys inside them.
CRYPT-KM-1	**KM rate**	Value: >400,000 · The key mover SHALL be capable of validating 400k keys per second.
CRYPT-KM-2	**KM encryption**	The key mover SHALL use AES-GCM encryption with a 256 bit key size and a 96 bit deterministic incrementing IV.
CRYPT-KM-3	**KM attribute validation**	The key mover SHALL include and validate all applicable attributes in the additional authenticated data.

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
CRYPT-KM-4	**KM caching and performance**	The key mover MUST support full rate device operation with every key involved being moved from the host. Firmware MAY cache keying material but MUST remain FIPS 140-3 compliant.
CRYPT-GCM-1	**IV handling**	In each AES-GCM key slot in addition to the plaintext key itself and any other attributes, there MUST be two additional 12 byte fields. For hardware handling, one of these is a 12 byte IV counter ('IVctr') interpreted as a 96 bit big endian nonnegative integer, and the other 12 byte field for additional data ('AD') interpreted as a 96 bit field.
CRYPT-GCM-2	**IV Init**	When the key slot is first populated, the IVctr field MUST be able to be initialized with either a fixed value or bits from an approved RNG.
CRYPT-GCM-3	**Legacy mode**	The non-FIPS compliant GCM mode where the 96 bit IV is supplied in the SQE MUST be supported at the measured submission rate.
CRYPT-GCM-4	**Firmware mode**	The HW and FW design MUST make it possible to calculate the IV per submission with firmware, with capability to refer to and update the IV context data in the key vault.

5 Cryptographic Subsystem (Pools / Sequencers / Hasher / RNG)

30 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
CRYPT-1	**HW acceleration**	All cryptographic primitive operations (hashing, symmetric encryption, and the fundamental RSA, PQC, and ECC operations) MUST be executed in hardware accelerators, not in control core firmware.
CRYPT-2	**Pooled submission**	the control core cluster SHOULD access accelerators and plaintext keys by submitting requests to programmable entities that complete the entire operation and mediate access to the key vault.
CRYPT-3	**Open parameters**	Tessera's mutable firmware, ROMs, silicon design, or any individual Tessera die MUST NOT contain any static keys, obfuscation keys, serial numbers, default IVs, or constant numbers over 64 bits in size in ANY built production firmware, ROM settings, RTL strappings, or default fuse settings whose values are not among or derived from values among the following exceptions: * under the control of Caliptra and attestable * evident in OCP-SAFE reviewed sources * derived or added by open or third party or foundry provided toolchains * the result of physical calibration, trimming, yield recovery * evidently physically separated from influencing Tessera's operation in any mode
POOL-1	**No bottlenecks**	Pool managers MUST accept commands and return completions to firmware control cores at the rate necessary to meet the system performance goals based on the accelerators they manage.
POOL-2	**No overlap**	Pool managers MUST not share any underlying hardware accelerator resources with other pool managers.
POOL-3	**No direct addressing**	Operations into the pool from control core firmware MUST not be required to specify which physical accelerator to use.
POOL-4	**Multi step operations**	Pool managers or subsidiary sequencers MUST support all multi-step accelerator operations that require preserving individual context in an accelerator.
CRYPT-GEN-1	**OpenSSL format**	Cryptographic hardware handling tenant data MUST align to the OpenSSL (big endian) byte ordering convention.
CRYPT-GEN-2	**KAT support**	All cryptographic functions in the device MUST have provisions for executing FIPS-140 compliant boot and runtime known answer tests, and the architecture MUST specify in detail how this is achieved.
CRYPT-GEN-3	**FW interactions**	Signing, encryption, and key release HSM flows SHOULD require the control core firmware to have fewer than 10 interactions with SoC hardware, a pool manager, or sequencer in the overall itinerary of any individual tenant operation.
CRYPT-GEN-4	**Plaintext access**	Tessera SHOULD be able to be configured for operation with only sequencers, pool managers, hardware accelerators, the security processor, and specifically privileged DMA engines having access to key vault regions of GSRAM where plaintext keys and equivalently sensitive engine context is stored.
CRYPT-GEN-5	**Context save/restore**	Tessera's hashing and encryption IPs handling tenant data SHALL support context save/restore to GSRAM.
CRYPT-PQC-1	**Key as seed**	For both ML-DSA and ML-KEM, accelerator commands to the pool MUST work with keys both stored in their native expanded form and as a random seed. Key generation/expansion from a seed is not counted in throughput performance requirement.

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
CRYPT-PQC-4	**Deterministic variants**	PQC algorithms MUST be provided in ‘deterministic’ and ‘nondeterministic’ versions. The hardware accelerators must support a ‘deterministic’ version of all operations and use this for KATs.
CRYPT-PQC-5	**‘External mu’**	ML-DSA signing MUST be able to use an external or internal SHAKE engine in a compliant manner to efficiently achieve whole document signing at the data throughput rate specified for host memory hashing.
CRYPT-RSA-1	**CRT and compact support**	RSA keys SHALL be supported in compact or expanded for CRT (Chinese Remainder Theorem) form.
CRYPT-RSA-2	**OpenSSL conversion**	CRT formats MAY follow different equivalent schemes but conversion to and from OpenSSL notation without control core FW visibility of private keys SHALL be supported in the pool manager.
CRYPT-RSA-3	**Key sizes**	All RSA operations SHALL be supported on 2048, 3072, and 4096 bit modulus sizes.
CRYPT-HMH-1	**Host hashing function**	The control core firmware MUST be able to specify a host buffer (in the form of a PRP or SGL list) for SHA2 or SHA3 hashing, with the resulting digest directed to a specified GSRAM location and receive a notification when the resulting DMA and hashing work is complete
CRYPT-HMH-2	**Host hashing rate**	Value: >3.2GB/sec · The host memory hashing throughput MUST be at least 3.2GB/sec
CRYPT-HMH-3	**Host hashing max directive size**	Value: 16MB · The max amount of data allowed to be hashed in a single command MUST be at least 16MB
CRYPT-HMH-4	**No size limit**	The hashing context MUST be able to be saved and restored to the hash engine, to allow hashing documents of unlimited size or ones that are streamed into the host.
CRYPT-HMH-5	**Threads**	Value: >32 · The hashing functionality MUST be able to make progress on at least 32 pending directives at one time. This is to avoid head of line blocking owing to a single VM.
CRYPT-HMH-6	**Quantum**	Value: <=4kB · The hasher MUST process no more than 4kB of data from any one pending command before making progress on another pending command.
CRYPT-HMH-7	**GDMA priority**	The hasher MUST have its own unique priority level to the GDMA engine.
CRYPT-HMH-8	**Concurrent perf**	Host hashing performance MUST be achieved simultaneously with the hashing performance required to keep up with mask generation operations needed for some public key signing and encryption operations.
CRYPT-RNG-1	**RNG compliance**	Tessera SHALL have a FIPS140-3 and NIST 800-90C compliant random number generation directly accessible to the control core cluster, pool managers, crypto IPs.
CRYPT-RNG-2	**RNG perf**	Tessera RNG SHALL have the performance needed to meet random number generation requirements of full rate ECC, PQC, and GCM operations that require random number generation, as well as for GCM IV generation.
CRYPT-RNG-3	**RNG**	All cores running firmware must be able to access RNG to sample entropy.
CRYPT-HYBRID-1	**Hybrid support**	Standard: 5.5 of SP800-227. · The architecture MUST support the flows in section 5.5 of SP800-227.

6 Root of Trust and Physical Security

29 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
ROT-1	**Caliptra 2.2**	Tessera MUST have a Caliptra 2.2 subsystem root of trust, isolated from the data path and control processors of the ASIC as the security processor
ROT-2	**Caliptra exclusively**	There SHALL NOT be any other root of trust on Tessera.
ROT-3	**Caliptra streaming boot and recovery**	Tessera MUST implement OCP secure boot, OCP recovery, and OCP streaming boot through one of Caliptra’s dedicated I3C interfaces, and this MUST be the basis of booting every processor on Tessera in production operational mode.
ROT-4	**Caliptra SPDM, DOE**	Caliptra v2.2 subsystem SHALL implement SPDM and MUST have exclusive access to the DOE interface.
ROT-5	**Caliptra fuse lifecycle, decommissioning**	Tessera’s Caliptra integration MUST support fuse zeroization of the UDS and Field Entropy for decommissioning.
ROT-5A	**Stable Device Identity**	Tessera SHALL derive stable device identity material from the Caliptra identity hierarchy such that firmware updates, warm resets, and operational lifecycle events do not change the device's attestation identity unless the underlying device identity root material is intentionally re-provisioned or zeroized.

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
ROT-5B	**Derived Identity Keys**	Tessera SHALL support deterministic derivation of attestation-related identity keys from Caliptra-managed root identity material and SHALL document the derivation hierarchy and lifecycle behavior.
ROT-6	**Attestation**	Caliptra SHALL attest all claims about the security relevant SOC configurations and properties of Tessera.
ROT-7	**Measurement**	Caliptra SHALL load and measure all firmware loaded on Tessera. Boot and runtime measurements SHALL be supported independently.
ROT-9	**Interface enabling**	Tessera SHALL support enable and disablement of the PCIe attestation interfaces.
ROT-10	**Caliptra integration**	Caliptra's integration guidelines MUST be adhered to, and all relevant Caliptra trademarks MUST be acquired.
ROT-11	**Anti-rollback**	Tessera's Caliptra integration MUST support an anti-rollback counter for non-Caliptra firmware with at least 32 counts.
ROT-12	**SAFE review of Caliptra integration**	An OCP-SAFE Scope 3 review of Tessera's Caliptra integration MUST be acquired.
ROT-13	**Immutable Ownership State**	Device ownership and lifecycle state information used for attestation, provisioning authorization, and ownership transfer decisions SHALL be protected against modification by non-Caliptra firmware and SHALL be derived from immutable or securely managed root-of-trust controlled state.
ROT-14	**Physical Unclonable Function**	Tessera SHALL incorporate a Physical Unclonable Function (PUF) that may be used as a source of device-unique entropy and identity material for the Caliptra root-of-trust subsystem.
ROT-15	**PUF Integration**	The architecture SHALL define how PUF-derived material, fuse-derived material, or a combination thereof is incorporated into Caliptra identity derivation and attestation flows.
CINTR-1	**Anti-tamper feature**	Tessera SHALL support 2 x tamper detection 16-bit RTC battery backed registers and input pins, powered from the RTC power rail.
CINTR-2	**RTC domain power**	Tessera's tamper pins SHALL be RTC-domain powered to enable intrusion detection when the main power to the system is not present.
CINTR-3	**Timestamp registers from RTC**	Tessera's SP SHALL have 2× 16-bit tamper detection timestamp registers powered from RTC
CINTR-4	**Two input pins for tamper events**	The device SHALL have 2x input pins for tamper detection events powered from RTC.
CINTR-5	**Only external coin cell power**	The ultra-low power (deep sleep mode) power mode powers only the RTC and BBRAM and input pins and the timestamp registers. Tessera's BBRAM input SHALL be powered from the external CR2032 battery and SHALL NOT use any alternative battery.
CINTR-6	**Tamper detection raises level interrupt to SP**	Tessera MUST allow tamper detection events after power restore to interrupt the SP, so that SP firmware is the first actor on system intrusion.
HW-MONO-1	**Monotonic counter**	Tessera SHALL support 1 × 63-bit free-running monotonic counter atat least 50ns granularity.
HW-MONO-2	**Counter config**	Tessera's monotonic counter SHALL be secured against modification beyond unit increment per clock tick.
HW-MONO-3	**Counter clock protection**	The clock that feeds Tessera's free-running counter SHALL be protected against modification except from a secure processor running signed firmware.
HW-MONO-4	**Counter reset**	Tessera's monotonic counter SHALL be resettable only upon cold boot, thermal fault that risks permanent device damage, or other environmental conditions that could lead to unreliable increment.
HW-DBG-3	**Fuse based JTAG and UART disable**	Tessera SHALL have the capability to lock down device interface with fuse-based controls to prevent leakage of confidential data.
PKG-3	**FIPS opacity**	To comply with FIPS 140-3 level 3, Tessera's package SHALL be opaque within the visible spectrum.
PKG-4	**FIPS**	Tessera's package SHALL be lidded and be tamper-evident or ensure attempts to penetrate/remove the die from enclosure shall have high probability of serious damage to the device causing non-function.

7 Reset, Timers and Interrupts

20 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
SYS-RST-1	**Multi domains**	Tessera SHALL support multiple reset domains.
SYS-RST-2	**Applicability**	It SHOULD be possible for firmware to hold any major piece of hardwired logic, subsystem or processor core in reset indefinitely and independently. General exceptions are listed here; other exceptions may appear within this requirements document: Functions that are RoT security related and intimately tied to the protection of device secrets. Free-running counter(s) that may be used for logging or global time-keeping services. Processor cores or logic that have been architected to perform as a unit or subsystem, and resetting one portion completely renders the jointly provided function inoperable.
SYS-RST-3	**SP reset keeps PCIe up**	The security processor (SP) subsystem SHALL permit reset of the security processor, independent of the reset of the CPU side PCIe endpoint. This enables live update and impact-less firmware update of the SP.
SYS-RST-4	**CPU reset does not cause SP reset.**	Reset/update of the CPU MUST NOT cause the reset of Tessera's root of trust security processor. The reset of the host application processors SHALL trigger PERST, whereby the device will clear key cache, PCIe queues and resources, protection and measurement and policies in the device shall be preserved.
HW-WDT-1	**Watchdog**	A watchdog timer SHALL be included with each CPU core on Tessera.
HW-WDT-2	**Watchdog persistence**	Tessera's watchdog flag SHALL be persisted across reset to permit detection of watch dog reset.
HW-WDT-3	**Watchdog expire time**	Tessera's watchdog flag SHALL contain counter value cleared on disablement of watchdog.
HW-WDT-4	**Watchdog interrupt**	Tessera's watchdog timer expiration SHALL generate maskable interrupt and reset signal which can be disabled by firmware.
HW-TMR-1	**Timers**	Tessera SHALL provide minimum 8 general purpose 24-bit interval timers that can generate interrupts.
HW-TMR-2	**Timer resolution**	Tessera's programmable timers SHALL have resolution under 10 ns
HW-TMR-3	**Timer interrupt to GPIO or core**	Timer interrupts SHALL be able to generate a pulse on a GPIO output or to on-device CPU core with firmware control.
HW-TMR-4	**Timer interrupt delay**	Each timer MUST be able to generate an interrupt a programmable number of clock ticks beyond an input hardwired event (active high, low, or pulsed) or a firmware write event.
HW-RTC-1	**BB domain**	Tessera SHALL support always-on battery backed voltage domain for RTC and intrusion detection.
HW-RTC-2	**BB oscillator**	Tessera SHALL support receiving independent oscillator which is on the always-on power domain
HW-RTC-3	**BB RTC**	Tessera SHALL support a 32-bit RTC counter (between 1 and 128 Hz) and match register for alarm generation
HW-RTC-4	**BB registers**	Tessera SHALL support a minimum of 2 × 16-bit registers that retain data in battery backed domain across cold power cycles.
HW-INT-1	**Interrupt controller**	Tessera SHALL include one (global) or more interrupt controllers.
HW-INT-2	**Programmable interrupt**	Tessera's interrupt controller SHALL support programmable interrupts.
HW-INT-3	**Interrupt destinations**	The interrupt controller SHALL be able to generate interrupts to multiple on-chip cores
HW-INT-4	**Interrupt types**	The interrupt controller SHALL support both level sensitive and edge triggered direct line interrupts

8 Peripheral IO

28 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
HW-QSPI-1	**QSPI**	Tessera SHALL support 2x Quad Serial Peripheral Interface (QSPI) to external Serial Flash NVRAMs with one interface connected to Caliptra and another interface available for the use of the control core cluster.
HW-QSPI-2	**QSPI chip selects**	Tessera's QSPI SHALL support 2x chip selects.
HW-QSPI-3	**QSPI boot and DMA**	Tessera's QSPI IP SHALL support programmable DMA from SPI flash FIFO to on-chip SRAM during ASIC boot. Internal security IP such as Caliptra will perform all authentication/integrity checking on the in-SRAM copy.

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
HW-QSPI-4	**QSPI size**	Tessera's NVRAM size supported SHALL be at least 250% of the all the memory space (TCM and optionally DTCM + other SRAM) allocated for all on-chip CPU core boot.
HW-QSPI-5	**QSPI speed**	Tessera's QSPI frequency SHALL be >= 33 MHz at 40pF signal output load.
HW-QSPI-6	**QSPI RPMC**	Tessera's QSPI controller for the control core cluster SHALL support replay protected monotonic counters (RPMC).
HW-I3C-1	**3x I3C**	Tessera shall support 3x programmable I2C/I3C interface (1x control plane, 1x BMC, 1x spare) with support for I3C 1.1.1 or newer.
HW-I3C-2	**I3C Telemetry**	I2C/I3C interface to BMC shall allow reading telemetry and critical Tessera information (temperature, critical errors etc.) without core cluster FW support (FW unavailability except Caliptra subsystem)
HW-I3C-3	**I3C speed**	I3C shall support 12.5 Mbps at 1.2 and 1.8V.
HW-I3C-4	**I3C OOB management**	I3C (BMC) shall support the requirements as specified in the PCISIG ECN Architectural Out-of-Band Management.
HW-I3C-5	**I3C/SMBUS support**	Shall support either SMBUS or IxC operations on a single interface. Dynamic reconfiguration is not required.
HW-I3C-6	**I3C interrupt support**	Shall support for I3C Inband Interrupts enabled from Tessera to BMC and Control plane
HW-I3C-7	**I2C fallback support**	I3C IP with support for I2C fallback.
HW-I3C-8	**SMBUS operation support**	Both initiator and target operations must be supported for SMBUS (dynamic reconfiguration is not required.)
HW-I3C-9	**NVRAM addressing support**	Configurable target addresses must be allowed to be programmed in NVRAM, but not limited by programmed values (see PCISIG ECN)
HW-USB2-10	**USB2 support**	USB2 device support for streaming boot, MCTP, OBMF with atleast 2 devices visible to USB host.
HW-USB2-11	**USB2 Support**	USB2 device 0 shall support streaming boot, MCTP
HW-USB2-12	**USB2 Support**	USB2 device 1 shall support MCTP & OBMF
HW-GPIO-1	**12 GPIO**	Tessera SHALL hall provide 12 General Purpose I/O pins.
HW-GPIO-2	**GPIO input default**	Tessera's GPIOs are inputs by default.
HW-GPIO-3	**1.8v/1.2v support**	Tessera's GPIOs SHALL support selectable 1.8V and 1.2V logic levels.
HW-GPIO-4	**12mA drive**	Tessera's GPIOs SHALL support programmable drive strength up to 12 mA.
HW-GPIO-5	**Debounce required**	Tessera's GPIOs SHALL have debounce or input glitch filtering capability.
HW-GPIO-6	**GPIO pullups**	Tessera's GPIOs SHALL support internal pull-up (PU) and pull-down (PD).
HW-GPIO-7	**GPIO interrupts**	Tessera's GPIOs SHALL support the generation of maskable interrupts to the SP and the core cluster via an interrupt controller, configurable as level (Hi/Lo) and edge (raising/falling).
HW-GPIO-8	**GPIO interrupts disabled at reset**	Tessera's GPIO interrupts SHALL be disabled at reset.
HW-DBG-1	**JTAG debug**	Tessera SHALL provide Serial Wire Debug capability or JTAG port to JTAG TAP controller(s).
HW-DBG-2	**UART**	Tessera SHALL provide a UART interface operable up to 115 kbps; software flow control option should be included.

9 Hardware Support for Firmware

15 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
FW-SVC-1	**Impactless update**	Tessera MUST implement impactless runtime firmware update for all its user programmable cores (cores related to IO's that necessarily impact PCIe link state are exempted). During updates, the host consumers of the PFs and VFs MUST still be able to submit I/O to all of their NVMe submission queues and without dropping any in-flight I/O. Tessera MAY undergo warm reset for an update to take effect while the OS is submitting requests to Tessera queues. The warm reset time must be less than 1 second.
FW-SVC-2	**PCIe controller up during updates**	The PCIe controller MUST be kept up to receive doorbells during the impact-less update operation.

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
FW-SVC-3	**Boot sources**	Tessera MUST be capable of booting from externally attached flash, although streaming boot will be used in production. For external flash boot, it MUST support at least three slots or partitions for each firmware image and configuration data to support recovery flows.
FW-SVC-7	**Update IO quiescing**	Tessera MUST support quiescing IO transactions during impact-less updates of device firmware. The device must be able to backup and restore head pointer, tail pointer and phase bit of NVMe queues.
FW-CDR-1	**Crash detection**	Tessera MUST be able to detect any core's non-operational state like hang or hard fault or memory access fault or intentional crash or bug check.
FW-CDR-2	**Cross signaling**	A core MUST have the ability to signal all other CPU cores when the fault occurs.
FW-CDR-3	**Crash dump collection**	Upon crash detection Tessera MUST have the ability to retrieve the crash dump information and save it in nonvolatile storage.
FW-CDR-4	**Configuration stable through recovery.**	Tessera crash recovery MUST retain the device configuration and PCIe link states.
FW-CDR-5	**Recovery reset**	Tessera crash recovery MUST hold all CPU cores and any stateful hardware IP (except PCIe) in reset.
FW-CDR-6	**Recovery memory scrubbing**	Tessera crash recovery MUST Wipeout instruction and data memory in the SoC.
FW-CDR-7	**Recovery reload**	Tessera MUST reload a working firmware version to all data and instruction memory
FW-CDR-8	**Recovery response**	Tessera's hardware and firmware must be in operational mode and responsive within 1 second of detecting a crash scenario.
FW-1	**Language**	All Tessera firmware MUST be implemented in Rust.
FW-10	**Open Source**	All Tessera firmware, toolchains, validation sources, and build system that are developed as part of the program MUST be published in open public domain for anyone's review.
FW-11	**Build**	All firmware compiler/build toolchains must support repeatable builds verification of firmware hashes against attestation.

10 Power, Thermal and Package

10 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
PWR-1	**Battery backed lifespan 2 years**	Tessera shall maintain battery backed power RTC and intrusion detection domains for a period of 2 years powered by a single 3V CR2032 battery.
PWR-2	**Power scenarios**	Tessera shall meet the TDP power ≤ 12 W with scenarios specified in the table below.
THERM-1	**Temp range**	Tessera's functionality SHOULD be validated across the operating temperature (die junction) range 0°C to 105°C, and supplier to report target T_case_max result with respect to 85 & 95°C die junction.
THERM-2	**Temp sensors**	Tessera's die SHOULD have one or more on-die temperature sensors with minimum delta_T offset between die hotspot and sensor.
THERM-3	**Simulated max hotspot**	Tessera SHALL provide maximum hotspot offset based on power/thermal simulations.
THERM-4	**Temp sensor accuracy**	Tessera's Temperature sensors SHALL have an accuracy level of +/- 4 degrees Celsius, with accuracy of +/- 2 degrees Celsius for the most important band of temperatures between 75C and 105C. Temperature sensors may need calibration during manufacturing. Calibration results and process to be provided to MSFT.
THERM-5	**Temp sensor USB**	Tessera's temperature readings SHALL be available to read out over USB, and I3C to accommodate legacy systems.
THERM-6	**Temp thresholds**	Tessera's temperature thresholds SHALL be provided as follows: * 1. E-Fuse values for minimum and maximum operating die temperatures that trigger a protective shutdown/reset mechanism. Fused values shall be retrievable from registers. 2. Programmable thresholds for minimum and maximum temperature alerts which include: 1. minimum operating temperature to initiate voltage ramp to increase die temperature. 2. critical operating temperature warning. 3. silicon power throttling temperature (if available, stepped response). 4. reporting temperature excursions under and over a certain programmed threshold via one or more IO.
PKG-1	**Dimension**	Package body (X, Y) dimensions for Tessera SHALL target 14 mm x 14 mm
PKG-2	**Type, pitch**	Tessera SHOULD be housed in a Flip-Chip BGA with a preferred ball pitch of 0.80mm and a minimum ball pitch of 0.75mm, subject to package body dimensions above

11 Manufacturing / DFT

12 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
MFG-KEY-1	**AKMS support**	Tessera's manufacturing SHALL support key management system and key harvesting and importation.
MFG-ATE-1	**Fault Coverage**	Tessera SHALL support ATE digital logic test via uncompressed scan, compressed scan or logic BIST. Fault coverage shall target > 99% DC/ Stuck-at Fault and 95% for AC/Transition Faults
MFG-ATE-2	**BIST**	Tessera SHALL support ATE internal memory test via BIST on all memories including latch-array, register file, and multi-ported memories, targeting 100% coverage.
MFG-ATE-3	**OTP**	Tessera SHALL support ATE OTP and/or eFuse test of the read and write logic.
MFG-ATE-4	**Signal**	Tessera SHALL support ATE Analog/Mixed Signal test.
MFG-ATE-5	**PCIe**	Tessera SHALL support ATE I/O test including PCIe.
MFG-ATE-6	**IO**	Tessera's high-speed IO (PCIe) IP SHALL enable at-speed PHY loopback at different levels: Internal, pad/bump level, and external (far-end loopback)
MFG-ATE-7	**Regulators**	Tessera's design SHOULD have bypass features for any on chip regulators that are present.
MFG-ATE-8	**Coverage**	Tessera's design SHOULD include features to enable high toggle coverage (>95%) during HTOL with limited pin access.
MFG-ATE-9	**LBIST**	LBIST or Functional testing, e.g., using on-chip SRAM SHOULD be supported for functional coverage for DPPM improvements.
MFG-ATE-10	**Shipping**	Tessera devices SHOULD be shipped in secure state, with OTP programmed for secure boot but any field entropy reserved for the device owner unprogrammed.
MFG-ATE-11	**Serial Number**	Tessera Serial Number encoding SHALL include batch and lot.

12 Reliability (RAS)

7 requirements

ID	REQUIREMENT	DESCRIPTION (CRD TEXT)
SYS-RAS-1	**All memories protected**	All of Tessera's memories must have some form of ECC protection.
SYS-RAS-2	**Single bit correction minimum**	All ECC protection in Tessera memories MUST have minimum single bit correction.
SYS-RAS-3	**Memory ECC error interrupt**	Tessera must report ECC multi-bit errors in GSRAM memory, cache, and memories within any accelerator hierarchies visible to the control core CPUs, to all the CPU cores through interrupts.
SYS-RAS-4	**Bus interface ECC error interrupt**	Tessera must detect parity or ECC errors at all of the initiator and responder interfaces of hardware IPs and report it to all the CPU cores via interrupts
SYS-RAS-5	**Error injection infrastructure**	Tessera must provide error injection infrastructure to each of the ECC and parity error points, and all other hardware faults, with independent control over the location and type of error.
REL-SER-1	**SDC FIT**	Qty: 50 FITS · Tessera shall suffer no more than 50 Silent Data Corruption FITS
REL-SER-2	**DUE FIT**	Qty: 250 FITS · Tessera shall suffer no more than 250 Detected Uncorrected Error FITS

13 Performance Requirements (all PERF)

Hybrid public key operation means that the PQC and traditional (RSA/ECC) accelerators must operate at their full specified rate concurrently. Furthermore, the AES and hashing engines are needed in most public key operations.
All RSA and ECC ops are assumed to be performed on the same pool of accelerators, so the performance requirements listed for RSA and ECC operations are expected to be attained individually, not concurrently. The same holds true of ML-KEM and ML-DSA operations on the PQC accelerators.

ID	OPERATION	TARGET	UNITS	NOTES (CRD TEXT)
PERF-1	RSA Sign 2048 CRT	100,000	OPS	CRT assumed provided in OpenSSL format.
PERF-2	RSA Sign 3072 CRT	32,500	OPS	
PERF-3	RSA Sign 4096 CRT	14,500	OPS	
PERF-4	RSA Sign 2048	28,500	OPS	Compact format
PERF-5	RSA Sign 3072	9,000	OPS	
PERF-6	RSA Sign 4096	3,750	OPS	
PERF-7	ECC Sign p-256	200,000	OPS	
PERF-8	ECC Sign p-256k	200,000	OPS	
PERF-9	ECC Sign p-384	100,000	OPS	
PERF-10	ECC Sign p-521	60,000	OPS	
PERF-15	AES 128 CBC/ECB/CTR/OFB/GCM	1,600	MB/s	Utility support
PERF-16	AES 192 CBC/ECB/CTR/OFB/GCM	1,600	MB/s	
PERF-17	AES 256 CBC/ECB/CTR/OFB/GCM	1,600	MB/s	
PERF-18	RSA Gen 2,048	> 1	OPS	Needs to be much better than FW as implemented in MbedTLS library.
PERF-19	RSA Gen 3,072	> 1	OPS	
PERF-20	RSA Gen 4,096	> 1	OPS	
PERF-21	ECDHE p-256, p-256k	200,000	OPS	UPKA perf ~ ECDHE and ECC
PERF-22	ECDHE p-384	100,000	OPS	
PERF-23	ECDHE p-521	60,000	OPS	
PERF-24	SHA 1	2,800	MB/s	
PERF-25	SHA 2-256	3,100	MB/s	
PERF-26	SHA 2-384	3,200	MB/s	Bulk hashing rates matched to 16KiB document size for signature throughput.
PERF-27	SHA 2-512	3,200	MB/s	
PERF-28	SHA 2-512/224	3,100	MB/s	
PERF-29	SHA 3-224, 256, 384, 512	3,200	MB/s	
PERF-30	SHAKE128	3,200	MB/s	
PERF-31	SHAKE256	3,200	MB/s	
PERF-32	ML-DSA-44	200,000	OPS	Key gen, signing, verifying
PERF-33	ML-DSA-65	100,000	OPS	Key gen, signing, verifying
PERF-34	ML-DSA-87	100,000	OPS	Key gen, signing, verifying – retain parity with ECC-384 for hybrid
PERF-35	ML-KEM-512	200,000	OPS	Key gen, encaps, decaps
PERF-36	ML-KEM-768	100,000	OPS	Key gen, encaps, decaps
PERF-37	ML-KEM-1024	60,000	OPS	Key gen, encaps, decaps

14 Cryptographic Algorithms (CRD requirement names)

AES CRYPT-AES-1 … CRYPT-AES-9	AES-ECB · AES-CBC · AES-CTR · AES-CFB · AES-OFB · AES-GCM · AES-KW · AES-KW2 · AES-KWPAD
Hash / HMAC / KDF CRYPT-HASH-1 … CRYPT-HASH-7	SHA1 · SHA2 · SHA3 · SHAKE · HMAC-SHA2 · HMAC KDF, counter mode · HMAC extract/expand KDF (HKDF)
RSA padding / MGF CRYPT-HASH-8 … CRYPT-HASH-11	RSA-PKCS 1.5 padding · RSA-PSS padding · RSA-OAEP padding · MGF1 calculation
RSA CRYPT-RSA-4 … CRYPT-RSA-7	RSAES-PKCS1-V1_5 · RSA-OAEP · RSASSA-PSS · RSASSA-PKCS-v1_5
ECC CRYPT-ECC-1 … CRYPT-ECC-5	E256 · E256K · E384 · E521 · ECDH key exchange
PQC CRYPT-PQC-2 … CRYPT-PQC-7	ML-DSA ops · ML-KEM ops · ML-DSA (44, 65, 87) · ML-KEM (512, 768, 1024)

Requirements in the CRD that state different values

BASE-10	**Processor Complex size**	At least 4x M7 or E6 class cores, at least 900MHz. At least 64kB instruction cache and 64kB data cache per core.
CPU-3	**Core cluster clock rate**	Quantity: At least 1 GHz · Each core SHALL have at least 1GHz clock rate.
SEC-PCIE-4	**Attestation**	Attestation of the PCIe Endpoint device shall support Identity and device measurement claims according to SPDM 1.5 pending approvals in the standard bodies.
SEC-PCIE-FEATURES-1	**CMA BaseAsymAlgo**	Tessera SHALL have hardware support for the following sign and verify (“BaseAsymAlgo”) operations specified within Component measurement and authentication (CMA/SPDM1.4): RSASSA-3072 ECDSA P-256 ECDSA P-384 ECDSA P-521

15 Open-Source Status (public TesseraHSM repositories vs CRD)

as of 2026-10-29

REPOSITORY	DESCRIPTION (README TEXT) · CONTENTS
tessera-specs `@ bcdb753`	Tessera - Integrated HSM Specifications (CRD, Command, DARC, Definition, Sealing, TMK, Deployment Models)
tessera-rtl `@ fc5e739`	RTL IP for Tessera HSM cryptographic accelerators. src/: aes, ecc, hmac, hmac_drbg, mldsa_mlkem, rsa, sha2 (incl. masked), sha3_shake
tessera-sw `@ 3281212`	Tessera Integrated HSM (TSRA) SDK is a modular, cross-platform software development kit (SDK) written in Rust. This repository is home to TSRA SDK, its simulator, and its OpenSSL Provider. api, crates, ddi, fw (core / pal / plat)
tessera-linux-driver `@ a43d313`	This directory contains the Linux kernel driver source code for the Tessera Integrated HSM device. tsra.ko
tessera-sidecar-scripts `@ d3292a9`	Builds ddi_sock_server into a bootable guest image, either as a Unikraft unikernel running under MSHV or QEMU/KVM, or on a minimal Linux kernel under QEMU/KVM. build.sh, configs, scripts
unikraft `@ 0a6044d`	Unikraft powers the next-generation of cloud native, containerless applications by enabling you to radically customize and build custom OS/kernels; unlocking best-in-class performance, security primitives and efficiency savings. github.com/TesseraHSM/unikraft · used by tessera-sidecar-scripts

Hardware functions: 1 RTL public · 6 partial · 9 not public · 2 external IP. Rules: **RTL public** = RTL in tessera-rtl covering the parameters listed in the CRD; **Partial** = RTL exists but covers only some of the listed parameters/algorithms; **Not public** = no corresponding RTL in the public repositories; **External IP** = the CRD specifies an external provider (Caliptra, CPU cores).

CRD FUNCTION	REQUIREMENTS	PUBLIC SOURCE	STATUS
ML-DSA / ML-KEM	`CRYPT-PQC-1..7`	tessera-rtl/src/mldsa_mlkem_all_levels	RTL public
RSA	`CRYPT-RSA-1..7`	tessera-rtl/src/rsa_4k_3k_2k	Partial
ECC	`CRYPT-ECC-1..5`	tessera-rtl/src/ecc_ecdsa_ecdhe	Partial
AES	`CRYPT-AES-1..9`	tessera-rtl/src/aes_all_modes	Partial
SHA-1 / SHA-2 / SHA-3	`CRYPT-HASH-1..4`	tessera-rtl/src/sha2_*, sha3_shake_all_modes	Partial
HMAC / KDF	`CRYPT-HASH-5..7`	tessera-rtl/src/hmac_256, hmac_512_384	Partial
RNG	`CRYPT-RNG-1..3`	tessera-rtl/src/hmac_drbg	Partial
PCIe endpoint / IDE / TDISP	`SEC-PCIE-1..8, CRYPT-IDE-1..2`	none	Not public
Command fetch / queues / QoS	`BASE-4..9, QOS-1..5`	RTL: none · tessera-sw fw/plat/uno: iic / oic drivers	Not public
GSRAM / Key Vault / apertures	`MEM-1..6, KEY-1..4, KEY-PIN-1..2`	RTL: none · tessera-sw fw/plat/uno: vault driver, key_vault crate	Not public
Key Mover	`CRYPT-KM-1..4, KEY-OFF-7`	none	Not public
Pool Manager / Sequencer	`POOL-1..5, CRYPT-GEN-3`	none	Not public
Host Memory Hasher	`CRYPT-HMH-1..9`	none	Not public
PUF	`ROT-14..15`	none	Not public
RTC / intrusion / physical protection	`CINTR-1..6, PKG-3..4`	none	Not public
Peripherals (QSPI, I3C, USB, timers)	`HW-QSPI-1..6, HW-I3C-1..9`	none	Not public
Control core cluster	`CPU-1..5, BASE-10`	external cores · tessera-sw fw/plat/uno targets Cortex-M	External IP
Caliptra 2.2	`ROT-1..13`	external (chipsalliance/caliptra)	External IP